Ysbyty Gwynedd Anaesthetic Guidelines

Central Venous Catheter Insertion

Type:
General
Keywords:
General, ICU, Practical Procedures and Equipment

Key Points:

- **Use ultrasound if time permits**
- **Verify wire prior to dilation**
- **Ensure wire removed**

Those inserting central venous catheters should be competent in all aspects of the technique, including (but not limited to) assessing/performing:

- Indication & contra-indications
- Consent
- Site & catheter selection
- Ultrasound use
- Seldinger & modified seldinger technique
- Verification of position
- Troubleshooting
- Specific recommendations

Asepsis

Central venous catheterisation should be performed in an environment that permits use of aseptic techniques. A standardized equipment set should be available for central venous access. An assistant should be used during placement of a central venous catheter. Full aseptic techniques and maximal barrier precautions (e.g., sterile gown & gloves, cap, mask covering both mouth and nose, and full-body patient drapes). A 2% chlorhexidine-containing solution should be used for skin preparation in adults, infants, and children. If there is a contraindication to chlorhexidine, povidone-iodine or alcohol may be used. Unless contraindicated, skin preparation solutions should contain alcohol.

Site

Catheter insertion site selection should be based on clinical need. An insertion site should be selected that is not contaminated or potentially contaminated (e.g., burned or infected skin, inguinal area, adjacent to tracheostomy or open surgical wound). In adults, selection of an upper body insertion site should be considered to minimize the risk of infection.

Catheter selection

Catheters should be matched to indication and have the minimum number of lumens that are realistically anticipated to be required. Catheter length should be determined by the insertion site and the adjustable flanges should only be used if the catheter is longer than optimal for correct placement. The following lengths of catheter should be considered as a rule of thumb

RIJ 12.5 cm,

LIJ 15cm.

Final length should be judged by position on chest radiography.

Ultrasound use

Static ultrasound imaging should be used in elective situations for pre-puncture identification of anatomy and vessel localization. Real time ultrasound should be used for guidance during venous access when either the internal jugular or femoral veins are selected for cannulation. In or out of plane techniques are acceptable- the operator must be competent in the technique they select as each offers certain advantages and disadvantages- but a combined technique is probably optimal. It is highly recommended that the wire is verified as being within the vein prior to dilatation.

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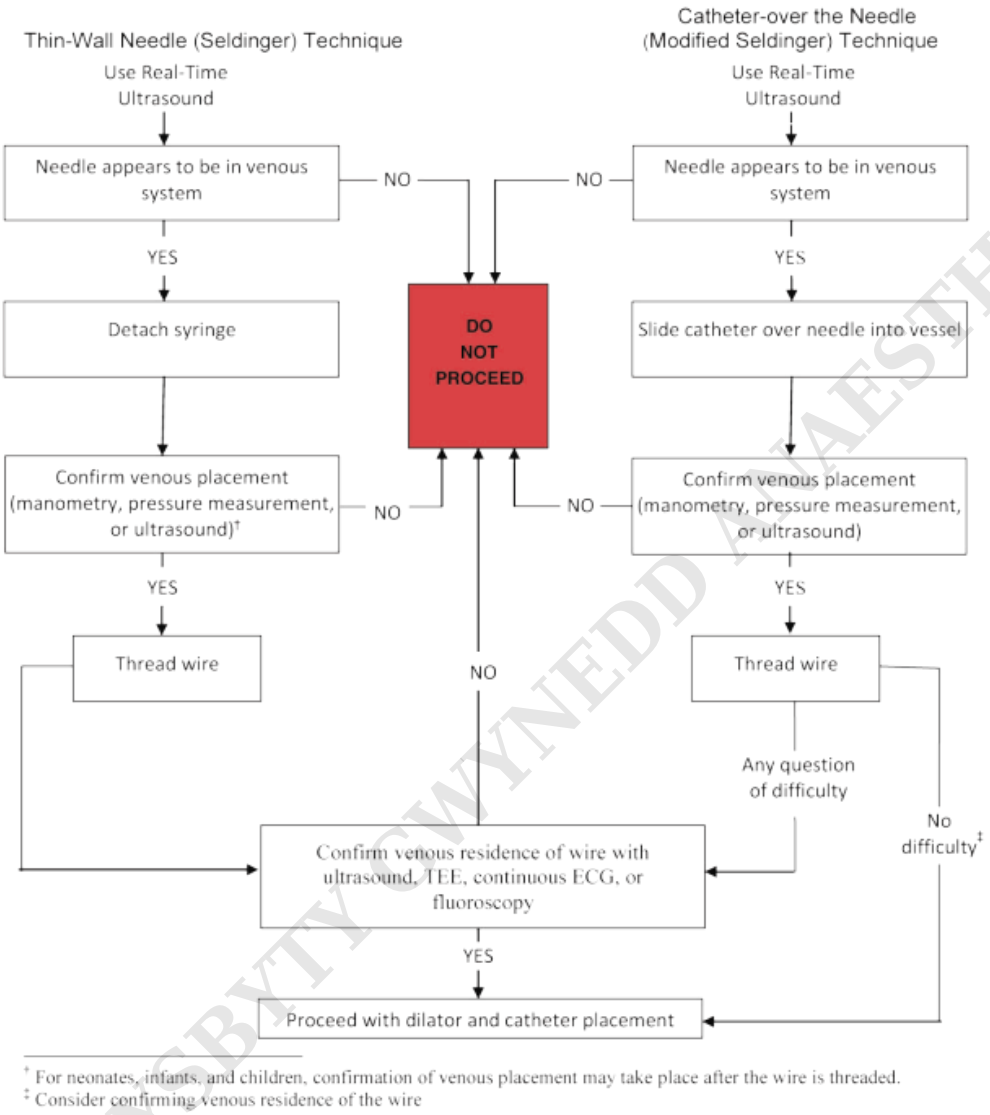
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Verification of position

Verification of position is a dynamic process as malposition may occur at any stage during insertion or following successful insertion. If, following an uneventful insertion (as indicated in the flowchart) a venous trace is obtained the line may be used if radiographic verification is likely to delay therapy. Radiographic verification should take place at the earliest possible opportunity.

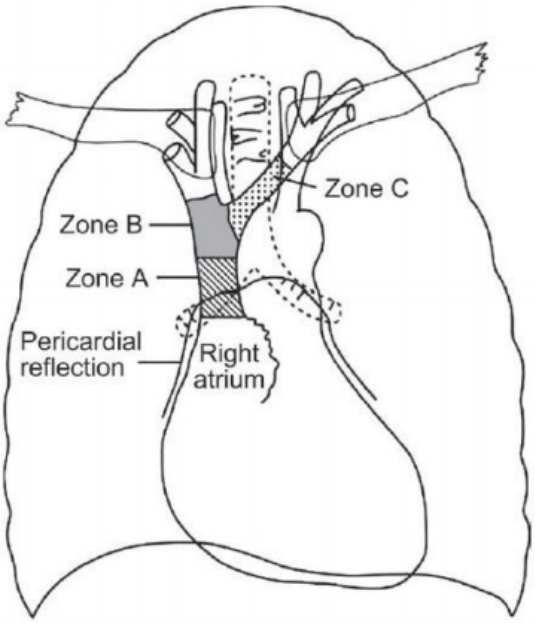
Algorithm for central venous insertion and verification.



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This algorithm compares the thin-wall needle (i.e., Seldinger) technique versus the catheter-over-the needle (i.e., Modified-Seldinger) technique in critical safety steps to prevent unintentional arterial placement of a dilator or largebore catheter. ***The variation between the two techniques reflects mitigation steps for the risk that the thin-wall needle in the Seldinger technique could move out of the vein and into the wall of an artery between the manometry step and the threading of the wire step.*** ECG = electrocardiography; TEE = transesophageal echocardiography.

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Correct placement is supported if the line follows the usual venous anatomy and the tip, lying parallel to the vessel wall is at or up to 2cm above the carina. Certain central catheters may be used if lying outside this position, but the decision to accept the position should be made on an individual basis by an operator experiences in such matters.

- Zones A, and B are acceptable if the angle of the catheter with the vessel wall is less than 40°.
- Zone C may be acceptable for short term fluid infusions, but should not be used for vasoactive or irritant drugs.
- There may be other acceptable placements for PICC and other such lines.

Malposition should be corrected as soon as possible. Arterial puncture with the needle or cannula is generally treated by direct pressure following removal of the needle. Intra-arterial catheter placement or dilatation of the vessel should be discussed with the vascular surgeons who will advise the correct course of action. Fixation In all cases, lines must be secured via the non-removable flange.

Care must be taken to avoid “cheese-wiring” of the suture through the skin by ensuring the suture is not too tight. If the catheter is longer than optimal for the selected insertion site, the additional, adjustable flange may be used, however the non-removable flange must also be secured to the patient with a suture.

Dressing

Transparent bio-occlusive dressings should be used to protect the site of central venous catheter insertion from infection. Unless contraindicated, dressings containing chlorhexidine may be used in adults, infants, and children. An absorbable dressing should be used if there is significant bleeding from the insertion site, but this should be replaced with a transparent, bio-occlusive dressing at the earliest opportunity.

References

Practice Guidelines for Central Venous Access American Society of Anaesthesiologists Anesthesiology, V 116 • No 3 539 March 2012
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Dr Simon Burnell
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